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User requirements for research data management services in Myanmar university libraries: A case study of the University of Yangon Library

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Abstract

University libraries in Myanmar are developing research data management services (RDMS) to support and boost the country's research and development. This paper reports on a research study that aims to identify and understand user requirements for RDMS. The University of Yangon Library (UYL) was selected as the research case, and 40 faculty members, research students, and library workers were approached and interviewed. The analysis of interview data adopted a thematic approach and used two analytic tools: coding for tagging and interpreting data; and concept map for visualising and demonstrating the progress of analysis. The analysis discerned 12 RDMS requirements in four theoretical themes: research data management planning, research data collection, research data processing and analysis, and research data preservation and sharing. Moreover, the analysis indicated that, to support the development of RDMS, libraries should enact three different roles, namely, participative role, educational and consultative role, and facilitative role. Also, practical implications were drawn from the analysis and suggested a three-stage action plan for RDMS development at the case library. Although this study focuses on libraries in Myanmar universities, the findings provide useful insights and implications for university libraries in other countries.

Keywords

case study, research data management service, user requirement, university library, Myanmar

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Introduction

University libraries are firmly positioned as the epicentre of university academic life and manifest the cultural capital and academic status of a university (O'Donnell and Anderson, 2022). Tenopir et al. (2015) state that, within the last decade with the growth of data-intensive research, the roles of university libraries have changed dramatically. University libraries and the academic research community have intensified their collaborative relationship (Baker and Powell, 2024; Koltay, 2016). Libraries now serve as research partners that are deeply involved in and contribute to all aspects of research projects, from research hypothesis articulation and data collection

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to publication and preservation of research output (Tenopir et al., 2019; Zhou et al., 2018).

To support data-centric research and respond to the increasing number of user requests, researchers and many university libraries globally have realised that the provision of research data management services (RDMS) is crucially important to diverse research activities (Adekoya et al., 2024; Cox et al., 2017). RDMS can be simply understood as a natural extension of traditional research support services provided in university libraries, although RDMS is more complex and comprehensive. Tenopir et al. (2012:7) defined RDMS as a systematic set of library services that address an entire range of data requirements along the full research lifecycle, "including data management plan, digital curation (selection, preservation, maintenance, and archiving), and metadata creation and conversion". As asserted by Whyte and Tedds (2011), RDMSs are centred on the organisation of data and concerned with how research data enter the research lifecycle and how to yield reliable and innovative results.

The status of RDMS development varies across countries and regions (Cox et al., 2017; Nwabugwu and Godwin, 2020; Ou and Zhou, 2016; Yidavalapati and Sinha, 2022). University libraries in North American and European countries are typically more advanced. Many have already designed and implemented RDMS to meet their own institutional needs and research priorities. In contrast, RDMSs are highly desired in many developing countries, but often have yet to be introduced or require further development.

The research project reported in this paper focuses on RMDS development in university libraries in Myanmar. Myanmar is located in the westernmost part of Southeast Asia between the two economic superpowers of China and India. With a population of about 54.38 million and a GDP of approximately 64.8 billion USD as of 2023, Myanmar is not one of Asia's most powerful and rapidly developing countries (Kemp, 2023; World Bank, 2024). The country has a long tradition of educational achievement and over 90% of the population is literate (Statista Research Department, 2023). Since the late twentieth century, the government of Myanmar has consistently emphasised educational development. At present, Myanmar has 164 higher education institutions and universities, including public and private universities, technical polytechnics, and vocational schools (Kaung, 2012; Swe and Yang, 2021). Myanmar's first university library was established in 1930 in the University of Yangon. In the 1950s, the arrival and support of Western library experts and educators led to the rapid development of several types of libraries, including those in universities (Kaung, 2012; Khurshid, 1970; Lwin, 2024).

Despite a reported lack of management and financial support (Kaung, 2012; Nyein, 2016; Oo, 2016; Wong and Kareng, 2024), libraries in Myanmar's universities are relatively well developed in terms of their technological and service infrastructures. Several researchers (e.g., Kaung, 2012; Oo, 2016; Swe and Yang, 2024) state that, in Myanmar, not only is there at least one library in each university, but the libraries also provide a variety of academic resources and services to support university teaching and learning, and user requirements relating to research and knowledge innovation (Kaung, 2012; Oo, 2016; Swe and Yang, 2024; Yee, 2002). Myanmar's university libraries have experienced a rapid and successful turn in the digital transformation with the nationwide implementation of the E-Library Myanmar Project, which digitises and interconnects all university academic resources across the country (Kupryte and Nyein, 2018; Nyein, 2016). Although some technological and managerial problems have yet to be resolved, the present digital infrastructure enables university libraries to advocate for changes, to fully utilise their roles in supporting learning and research, and to provide services of higher quality (Glutting et al., 2022; Kupryte and Nyein, 2018). However, RDMS presents a completely new set of challenges that require systematic changes, renovations and development in the existing library structure and service model.

Building upon the technological infrastructure and systems, the University of Yangon Library (UYL) has been actively seeking strategies and establishing programs to reinforce its central role in the university regarding research and information support. Developing an RDMS has been recognised and envisioned as a key project, planned and expected to be accomplished by 2025. Moreover, considering the pioneering positions of both the University of Yangon and its library, this RDMS project could lead, facilitate and encourage RDMS research and development in Myanmar.

This research aims to identify, qualify and understand the user requirements of RDMS at Myanmar's university libraries. Researchers assert that identifying and understanding user requirements are important

and should be an initiating step for the development of library RDMS (Li et al., 2020; Tenopir et al., 2019; Yidavalapati and Sinha, 2022). Many research studies have investigated RDMS requirements in North America (e.g., Averkamp and Gu, 2012; Joo and Peters, 2020; Lee et al., 2024; Tenopir et al., 2019), Europe (e.g., Arias-Coello et al., 2018; Martin-Melon et al., 2023; Senft et al., 2022). There is an increasing number of studies investigating RDMS requirements at university libraries in Asia, but mostly in China (Liu and Ding, 2016; Si et al., 2015; Yan et al., 2024) and India (R and Gala, 2021; Tripathi et al., 2017). There is virtually no research performed in Myanmar. As this project focuses on one specific university only – the University of Yangon – the user requirements are expected to be identified and comprehended from the perspective of faculty members, researchers, students, university management, and librarians. To guide the research design and the collection and analysis of data, the following research questions were formulated:

RQ1: What are the user requirements for RDMS at the University of Yangon Library?
RQ2: What implications can be drawn?
RQ3: How can RDMS requirements be transformed into library development strategies?

Literature review

RDMS at university libraries

In today's data-driven world of science, research data are central to nearly all scientific exploration (Nahotoko et al., 2024; Tripathi and Pandy, 2018; Webber and Kranzlmuller, 2019). The definition of research data differs from field to field. Broadly speaking, it refers to factual records in the forms of numbers, symbols, text, images and/or sounds that validate the research process and findings (Adika and Kwanya, 2020; Jiao et al., 2022; Si et al., 2015). Research data mainly include, but are not limited to, raw data, processed data, software, algorithms, protocols, methods, and materials (Elsevier, 2024).

According to Yoon and Schultz (2017), research data management (RDM) should be treated as an integral component of research activity and should not be separated from research. RDM, however, can be particularly complex because it is always deeply integrated with research processes within the research lifecycle. RDM positively supports scientific investigations because it can effectively promote research transparency, enhance research efficiency, ensure reliable verification of results, and allow future research to be built on existing findings (Adekoya et al., 2024; Donner, 2022; Patel, 2016; Tripathi et al., 2017). Pinfield et al. (2014) pointed out that RDM can be challenging because of the range of stakeholders in a university. Whilst RDM providers may include library workers, IT technicians, publishers, and data suppliers, service users usually include faculty members, researchers and students. Moreover, it is increasingly common for universities to develop research datasets collaboratively with external business partners.

Realising the importance of RDM, university libraries have been inspired to explore strategies to enhance their leadership role in RDM (Amanullah and Abrizah, 2023; Tenopir et al., 2012). Flores et al. (2015) assert that RDM presents both new opportunities and challenges for university libraries. It can be a great opportunity for university libraries to reinvent themselves, embrace new – and possibly more important – roles in a university, and become a real engine for research and innovation. Nwabugwu and Godwin (2020) claim that providing RDMS is critically important for university libraries. They advocate that university libraries should prioritise RDMS as essential and effective means to improve a library's visibility, deliver stronger impact on research activities in the university, and align library works and services with the university mission. Fearon et al. (2013) and Tunmibi and Ajokotola (2024) consider RDMS as a type of university library service that provides research data and related services that include research data consultation, training, planning, curation, sharing and publication. Moreover, RMDS are usually provided based on close and collaborative relationships between RDMS librarians and research teams (Amanullah and Abrizah, 2023; Brochu and Burns, 2019; Tang and Hu, 2019).

User requirements for RDMS

When establishing RDMS at a university library, it is fundamental to identify and understand user requirements and expectations (Gowen and Meier, 2020; Henderson and Knott, 2015; Nhendodzashe and Pasipamire, 2017). Through systematically consulting with users and understanding future users' requirements, university libraries can effectively define service priority and articulate effective strategies (Li et al., 2020; Maguire and Bevan, 2002). Previous studies have investigated user requirements for

RDMS, leading to important and useful findings (Cox et al., 2017; Si et al., 2015; Tenopir et al., 2012, 2015; Yidavalapati and Sinha, 2022). Table 1 shows a summary of RDMS requirements reported in the literature.

RDMS are now widely provided in university libraries around the world and have become essential and indispensable in research-oriented universities (Ashiq et al., 2020; Chiware and Becker, 2018; Cox and Pinfield, 2014; Safdar et al., 2023). Henderson and Knott (2015) and Lee et al. (2024) argue that data management training and data reference services are more requested than other RDMSs and thus need to be prioritised. Through the investigation of Chinese university libraries, Si et al. (2015) pointed out that librarians' direct involvement and participation in research projects is highly desired across various research groups and scientific communities. Lassi et al. (2016) asserts that RDMS should be aligned and provided throughout the lifecycle of research projects, accommodating a wide range of requirements that may arise at any stage of a project.

Despite previous efforts in identifying and understanding RDMS requirements, many studies (e.g., Tenopir et al., 2019; Tunmibi and Ajokotola, 2024; Yidavalapati and Sinha, 2022) report that RDMS requirements remain insufficiently understood and addressed. They point out that identifying the requirements is simply not enough. Rather, transforming the requirements into effective and satisfying user services is likely to bring greater challenges, which include but are not limited to raising users' awareness, building a capable service team, and maintaining sustained management and financial support. Therefore, Si et al. (2015) suggest that identifying, validating, understanding, and finally incorporating user requirements into library services should be continuous and inseparable steps in RDMS development.

Furthermore, the level of success in implementing and developing RDMS in university libraries around the world is inconsistent (Tang and Hu, 2019) for several reasons, including a lack of service mechanisms (Liu and Ding, 2016), inefficient technical infrastructure (Donner, 2022; Gonzalez and Peres-Neto, 2015), lack of supporting policies from the university (Si et al., 2015), limited skill and capability development of librarians (Chiware, 2020; Latham, 2017), and researchers' reluctance and unwillingness to share research data (Ismail et al., 2022).

Table 1. Summary of RDMS requirements reported in the literature.

Themes	RDMS Requirements	Sources
RDM planning	Consultative-type services for RDM planning	Amanullah and Abrizah (2023); Sinha et al. (2023); Safdar et al. (2023); Joo and Schmidt (2021); Masinda et al. (2021); Tenopir et al. (2012)
	RDM planning training	Kinde et al. (2021); Masinda et al. (2021); Mulongo (2021)
	Librarian participatory support in RDM planning	Hombali (2022); Peters and Dryden (2011)
	Provision of tools for RDM planning	Hombali (2022); Berman (2017); Yu (2017); Jones et al. (2013)
Research data collection	Librarian participatory support on research data discovery and collection	Yidavalapati and Sinha (2022); Zhou (2018); Martin-Melon et al. (2023)
	Institutional research data repository	Amanullah and Abrizah (2023); Bishop et al. (2022)
Research data processing and analysis	Tools for research data processing and analysis	Amanullah and Abrizah (2023); Miller et al. (2018); Lassi et al. (2016)
	Data processing and analysis training	Martin-Melon et al. (2023); Masinda et al. (2021)
	Librarian participatory support in data processing and analysis	Amanullah and Abrizah (2023); Joo and Peters (2020); Miller et al. (2018); Lassi et al. (2016)
Research data preservation and sharing	RDM technological infrastructure	Martin-Melon et al. (2023); Masinda et al. (2021)
	Long-term preservation strategy	Bishop et al. (2022); Yidavalapati and Sinha (2022); Li et al. (2019)
	Strategy for secured data sharing	Safdar et al. (2023); Masinda et al. (2021); Li et al. (2020)
	Research and data publications	Amanullah and Abrizah (2023); Yidavalapati and Sinha (2022)
	Risks and violations prevention	Donner (2022); Safdar et al. (2022); Yidavalapati and Sinha (2022); Li et al. (2020); Singh et al. (2018); Tripathi and Pandy (2018)

An extensive search and review of the literature revealed some articles on the development of university libraries in Myanmar. However, no research has focused on RDMS development in Myanmar's university libraries. The literature review showed that Myanmar's university libraries are still struggling with developing comprehensive academic collections and digital resources, implementing sophisticated technological infrastructure, updating IT solutions that are described as "currently primitive", and finding solutions for shrinking budgets and rising costs (Bradley, 2017; Nyein, 2016; Swe and Yang, 2024). In this case, implementing RDMS seems to be challenging and complex – the present library operation and IT infrastructure may not be supportive, and internal and external barriers preventing the adoption of RDMS are also evident (Oo, 2022). Nevertheless, it is difficult to find more information and evidence from the literature.

RDMS development in Myanmar university libraries

The Government of Myanmar has demonstrated ambitions to develop its own research infrastructure, as outlined in the National Education Strategic Plan 2016–2021 (NESP1) and to be expanded in the framework for NESP2 2021–2030 (Glutting et al., 2022; Ministry of Education, 2024). Moreover, the government is establishing the "Centre for Research and Innovation" to demonstrate a strong motivation to rejuvenate Myanmar's academic research, which is promoted as a vehicle to inform practice, learn, and establish cross-sector partnerships (Lee et al., 2020).

Nevertheless, despite concerns and attention on developing RDM infrastructure and schemes, research data remains largely controlled by foreign entities. Glutting et al. (2022) note that almost all social science research data generated in Myanmar are through foreign-led projects and donor-funded programs, a fairly typical situation in developing countries. Therefore, it is hoped that Myanmar's Higher Education sector can lead research data development and create opportunities for government agencies, research institutions, and foreign entities to work together to reinforce the development of indigenous RDM.

For Myanmar university libraries, RDMS is envisioned as a strategic development that can enhance library visibility and its central role in the academic life of university faculty, researchers and students (Missingham, 2017; Nyein, 2016; Swe and Yang, 2024). As informed by the literature, RDMS development is of pivotal importance to university libraries primarily from two perspectives (e.g., Nwabugwu and Godwin, 2020; Singh et al., 2022; Zhou et al., 2023). First, through providing RDMS, libraries have closer and stronger relationships with research teams and become essential for data-intensive research and the development of the latest information technologies, such as artificial intelligence technologies. Second, RDMS offers opportunities for continuous education that keeps researchers and students updated and equipped with the most advanced RDM concepts, platforms and tools.

However, although providing RDMS at university libraries in Myanmar may sound very appealing, it is yet unclear how to develop, operationalise and maintain these services. Also, it is yet to find out what the potential challenges and risks are, as well as potential enablers and facilitators. The literature review provides two important indications. Firstly, the existing research and literature cannot provide adequate answers to the research questions of this project. There is a significant lack of research on developing university libraries in Myanmar. Secondly, although there is a lack of directly relevant studies, the research team through reviewing the literature developed a knowledge background, which can offer useful and necessary theoretical underpinnings for this project.

Research methods and design

Research design

To effectively respond to the research questions, this study adopted an inductive qualitative approach and included two inductive stages: a literature review, and a case study.

The literature review was carried out between January and June of 2023 aiming to develop a conceptual background about what is already known and of knowledge gaps and deficiencies that require further investigation. The literature review followed a two-stage design. A generic review was undertaken in the first stage to understand RDMS development in university libraries around the world. For this stage, two academic databases: Web of Science, and Scopus, were searched for relevant studies and articles. The second stage of literature review was more concentrated on RDMS development in Myanmar. At the beginning, the research team searched the websites of Myanmar's university libraries. The search, however, showed limited information regarding RDMS. Thus, the research team decided to look for

information from academic databases. Specifically, the MERAL Portal (Myanmar Education Research and Learning) database was searched using the following keywords: "research data" OR "data service*" or "research data management". However, the search returned only one relevant research publication, suggesting the need to adopt an exploratory research design. The literature review provided an overview of general RDMS requirements (as summarised in Table 1) that served as a basis for the case study investigation.

Case study is a widely used research approach in social sciences and in the discipline of library and information science in particular (Fidel, 1984; Ngulube, 2022; Radford and Kern, 2006; Ylikoski and Zahle, 2019). In this study, a single case design was considered as appropriate and chosen for two main reasons: there is very little research reporting on the development of university libraries in Myanmar, and thus, limited information on RDMS in Myanmar's university libraries.

This study employed the University of Yangon Library (UYL) as the case study. The University of Yangon is the number one ranked and best-known university in Myanmar with a strong international reputation (Nyein, 2016). UYL, established in 1927, is the oldest and largest university library in Myanmar (Kaung, 2012). The library is famous for its collections of ancient scripts (palm-leaf bundles, handwritten manuscripts, ink stone inscriptions) from Myanmar and other countries in Southeast Asia, and for the provision of an impressive range of high-quality academic e-resources (Pyone, 2016). Aimed at supporting all research and teaching activities at the university, UYL has become an indispensable component of the university's research and innovation, and for students' everyday life. This research was commissioned by UYL. Thus, selecting UYL as the research case was both natural and desired. The leadership of UYL guaranteed administrative support and allocated resources needed for achieving the objectives of the project, including finding appropriate interviewees for the case research. Given that many of Myanmar's university libraries are attempting to establish RDMSs, the UYL project is expected to become a "showcase" from which other university libraries can learn and seek insights and guidance.

Interview data collection

The case study adopted an explorative qualitative approach, which was operationalised using a semi-structured interview and a thematic analysis. The design of the semi-structured interview questions was mainly based on the RDMS requirements identified in the literature. Specifically, RDMS requirements in Table 1 were converted into two sets of open-ended interview questions, one for the researchers, and the other for the librarians. The interview questions are shown in Table 2 below.

It is necessary to note that the research team received administrative and financial support from the home institution for a data collection trip to Myanmar for no more than 45 days. Considering the financial and time limitations for data collection, as well as the ongoing social and political instability in Myanmar, the research team opted to formulate a strategy that could allow for collecting as much data as possible and as quickly as possible. Therefore, a convenience sampling strategy was employed. According to Etikan et al. (2016), a convenience sampling strategy is sometimes regarded as an "accidental" strategy, because participants selected in the sample just simply happen to be "situated, spatially or administratively, near where the researcher is conducting the data collection". In this case, researchers select subjects who are more readily accessible and accept that not all qualified individuals in the target population have equal opportunity to participate in the research (Etikan et al., 2016).

During the field trip, the research team received valuable support from the UYL management team. They assisted in identifying suitable faculties and staff, who were then invited for an interview by the research team. The students interviewed were approached and invited by the research team by asking if they were research students, and if they were interested in RDMS. The research team managed to interview a total of 40 participants. As shown in Table 3, the interview sample includes 14 faculty members, 11 librarians, 2 tutors and demonstrators, and 12 research students.

The sample size of 40 was the maximum the research team could possibly reach. According to Morse (2000), qualitative research should be more concerned with the quality of data, rather than the size of the sample and the number of interviews. Having a large interview sample cannot guarantee that the data is of high and expected quality (Morse, 2000). In this research, instead of achieving a higher number of interviews, the research team decided to focus on the quality of each interview and the collection of meaningful data from each participant. To achieve so, three strategies were employed. Firstly, the design of interview

Table 2. Interview questions.

Categories	Researchers/Users	Librarians
Generic questions	What is your role at the University of Yangon? What types of data do you usually use and collect?	What is your role at the University of Yangon Library, and how do you assist researchers with their research data requirements?
RDM	What is your understanding of research data and how do you manage your research data?	How do you define research data, and what do you think important for effective RDM?
	Have you received any support for RDM from the library? If so, what did you find helpful? If not, what kinds of RDMS do you need?	What kinds of RDMS does the library currently provide?
RDM planning	What is the significance of developing a RDM plan, how do you create it?	Why is it important for researchers to develop a RDM plan?
	Would you find it beneficial to receive support in formulating your RDM plan, and if so, how would you like that assistance to be structured?	What kind of assistance do you think researchers might need when developing their RDM plans?
Research data collection	How do you collect and evaluate research data for reuse?	What approaches do you suggest researchers for collecting and evaluating their research data for reuse?
	What kinds of support do you expect from the library regarding research data collection?	What kinds of request have you received from researchers regarding research data collection?
Research data processing and analysis	How do you store and process your research data?	Could you describe any resources or practices provided by the library for processing and analysing research data?
	How do you explore and interpret the processed data, and what kinds of support do you expect from the library regarding research data processing and analysis?	What kinds if support do you offer to users on exploring and interpreting their processed data?
Research data preservation and sharing	How do you ensure the long-term preservation of your research data?	What approaches do you suggest for researchers to ensure the long-term preservation of their research data?
	Have you shared or published your research data, and if so, what challenges have you faced?	What challenges do you notice when researchers share or publish their research data?
	What kinds of support do you expect from the long-term preservation or sharing of your research data?	How does your library support the researchers to publish or share their research data safely?
Final questions	What kinds of RDMS do you expect to gain from the University of Yangon Library in the future?	What kinds of RDMS do you think the library should offer?

questions (as shown in Table 2) was based on RDMS requirements identified in the literature (as shown in Table 1). Secondly, the interview question script was trial tested with seven research students at the base institution. Thirdly, apart from the predefined interview questions on the question script, the research team anticipated and prepared trigger questions and follow-up questions during the interviews.

The demographic information of the interviewees is shown in Table 2.

Each interview lasted 20–45 min. With the participants' consent, all the interviews were recorded and then manually transcribed in Word documents.

Interview data analysis

The analysis of interview data employed a thematic analysis method. Two thematic analysis tools were used: coding and theoretical representation, as suggested by Zhou et al. (2021). The analysis of the qualitative data used a qualitative data analysis application, MAXQDA 2022, to examine, interpret, and constantly compare emerging codes. A screencap of MAXQDA analysis is shown in Figure 1 below.

The interview transcripts were coded using three types of coding: open coding, axial coding, and selective coding, as proposed by Strauss and Corbin (1998). Open coding examined, identified and labelled

meaningful data segments, whereas axial coding tried to interrelate each open code and interrelationship among them, through which processes themes and sub-themes emerged. Selective coding begins after the identification of potential themes. It is useful to identify the core findings by integrating relevant concepts and identifying what is missing or incorrect by checking and validating the findings against the raw data.

Theoretical representation is the use of a concept map and theoretical narrative to report the research findings. A concept map was developed and evolved throughout the analysis, to visualise and demonstrate the process of coding and the interrelationships between concepts. A synthesised theoretical narrative is a qualitative report of the research findings, presented in the following section (Table 4).

As shown in Table 3, the analysis of interview data pointed to 12 RDMS requirements, which emerged in four main themes: RDM planning, research data collection, research data process and analysis, and research data preservation and sharing.

Table 3. Demographic profile of participants.

Background	Number of participants
Gender	
Male	4
Female	36
Professional Position	
Professor (Head)	1
Professor	2
Assistant Professor	4
Librarian	2
Assistant Librarian	9
Lecturer	5
Assistant Lecturer	2
Tutor/Demonstrator	3
Research Students	12
Total	$N = 40$

Research findings

This section presents the research findings that emerged from the analysis of interview data. Specifically, this section presents and discusses the

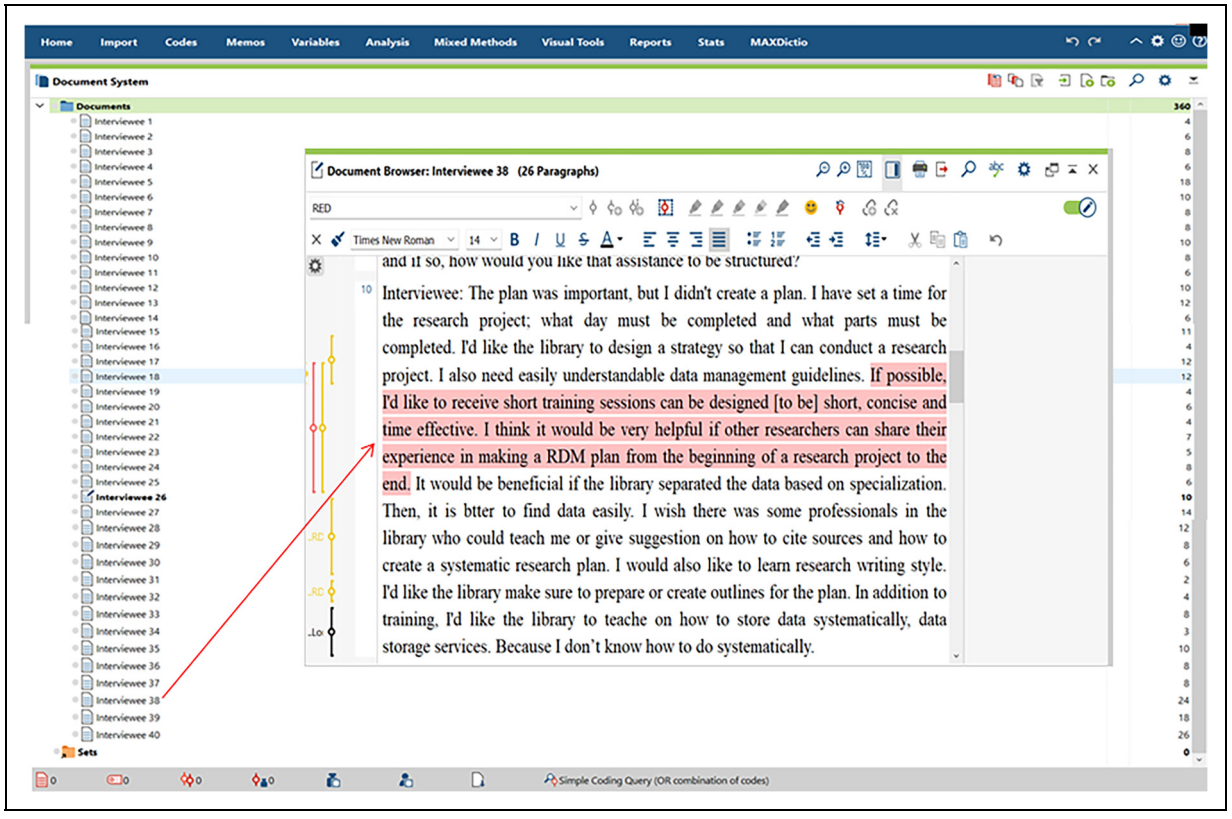


Figure 1. A screencap of MAXQDA analysis panel.

Table 4. RDMS requirements, main themes and supporting quotations.

Main Themes	Code ID	Requirements	Samples of supporting quotations
RDM planning	P1	RDMS department and advisors	"I wish there was some professionals in the library who could teach me or give suggestion on how to cite sources and how to create a systematic research plan" (Interview 38).
	P2	Comprehensive RDM training arrangement	"It is necessary to manage the research data systematically of good projects. I wish the library can organize related training workshops to offer planning strategies for those novices in managing research projects, such as where to start, how to start, and what must be done." (Interview 40).
	P3	Specific and explicit RDM guidelines	"I always seek advices from experienced professors and the literature. But I would really appreciate it if the library could provide a full-scale RDM guideline" (Interview 36).
Research Data Collection	C1	Access to open data resources	"E-access to online and open data would be very useful. I prefer an open platform for research data resources" (Interview 38).
	C2	Guidance on research data collection	"I am seeking guidance from the library on how to find and collect research data, as well as from which source I can gain research data. It is essential for me that the library's research data is easily accessible" (Interview 38).
	C3	Institutional research data repository	"I hope the library can set up a dedicated storage server to backup, store, and preserve research data permanently. If the data storage services are established by the library, I would be interested in using them" (Interview 40).
Research Data Process and Analysis	PA1	Training on research data processing	"I think our library should provide a lot of data processing support. Most researchers are processing their data by self-learning. The library can help a lot in this aspect" (Interview 36).
	PA2	Access to data analysis applications	"I only use Microsoft tools for data analysis. I want to know and learn software and applications that are easy and cheap to use for research data analysis" (Interview 8).
	PA3	Hand-on-hand assistance with research data processing and analysis	"If the library can help some parts of my data analysis and interpretation, it would be tremendously helpful" (Interview 35).
Research Data Preservation and Sharing	PS1	Long-term preservation of research data	"Sometimes, I save my data on the D drive and a USB to avoid damage. But the library does not provide any information about data storage or long-term preservation" (Interview 12).
	PS2	Promoting awareness about the importance of data preservation and sharing	"My colleagues and I are eager to learn more effectively methods to store and maintain our research data" (Interview 17).
	PS3	Policies and regulations to encourage research data sharing	"Regarding the publishing and sharing data, I really hope the library can introduce clear rules and protocols" (Interview 5).

RDMS requirements identified and thus responds to the first research question (RQ1). Responses to RQ2 and RQ3 are discussed later in Section 5, because they are developed based on the research findings.

According to the interview data, the university faculty members and researchers consider the library's RDMS "absolutely important" (Interview 5, 19, 36), "essential" (Interview 2, 3), and "necessary" (Interview 1, 12). From the perspective of the interviewees, UYL has a critical role in the provision of RMDS, where "personalised guidance and suggestions can be constantly provided" (Interview 12) and where "people know how [and where] to ask for help" (Interview 8). The development of RDMS at UYL should effectively respond to "the user's expectations" (Interview 1, 2, 18) and quickly answer "RDMS-related questions and requests raised by researchers" (Interview 22). This means that identifying user requirements is an unavoidable step

for RDMS development. However, presently UYL is not serving as an active research support that is deeply and actively involved with research projects and activities. It was observed when collecting data in the field that UYL was most likely not ready for RDMS.

Identifying user requirements for RDMS is important and indispensable for RDMS development. In this study, the analysis of 40 interview transcripts pointed to 12 RDMS requirements, which emerged in the following four themes. Correlating the findings with the literature review, it can be confirmed that the identification of four themes is supported by previous studies. The four themes are:

- RDM planning: makes sure all potentially required RDMS and demands are achieved, from research project inception to publishing and archiving research results (e.g., Ball, 2012; Gajbe et al., 2021; Si et al., 2015).
- Research data collection: provides necessary help and assistance to researchers in not only creating and collecting new research data, but also locating, acquiring and assessing existing secondhand data (e.g., Rousi, 2023; Tunmibi and Ajokotola, 2024).
- Research data processing and analysis: involves a variety of RDMS that support data preparation, analysis, process and interpretation, as well as that draw implication from data (e.g., Ball, 2012; Rousi, 2023; Zhou et al. 2018;).
- Research data preservation and sharing: refer to RDMS that stores, prepares, and plans research data for long-term preservation, as well as ensures data availability and re-use (e.g., Ball, 2012; Si et al., 2015; Zhou et al. 2018).

RDM planning

Making an RDM plan for managing research data is "vital" (Interview 18), "mandatory" (Interview 6, 9), and "extremely useful, essential, and effective" (Interview 3, 5) in a research project. A research project "can simply go wrong without a systematic data management plan" (Interview 4). The analysis of interview data pointed to three RDMS requirements at the planning stage:

- RDMS department and advisors
- Comprehensive RDM training arrangement
- Specific and explicit RDM guidelines.

It is natural for UYL to serve as an RDMS provider, advisor, and educator. As reflected in the data analysis, most of the interview participants need "support from the university library to … create a good and workable data management plan" (Interview 2, 17, 36). Researchers expect "the library can turn into the ultimate RDM planning advisor", where there is "always someone who can guide and give suggestions for the RDM plan" (Interview 12). Thus, there is clearly a need to have a dedicated RDMS department within UYL, where a professional team of advisors can offer a range of services.

Moreover, researchers expect UYL to provide comprehensive training services in forms such as lectures, seminars, and workshops. These training sessions should focus on developing theoretical knowledge, and demonstrate "how to use [relevant] software, applications, and platforms to create an RDM plan" (Interview 14), as well as sharing "experience of successes and losses" (Interview 37). Nevertheless, researchers currently receive very limited help and assistance because there are "no training programs and sessions" (Interview 38).

> If possible, I'd like to participate in short training sessions, but I hope the sessions can be designed [to be] short, concise and time effective. I think it would be very helpful if other researchers can share their experience in making a RDM plan from the beginning of a research project to the end. (Interview 38)

> About 2 years ago, when the library bought the EBSCO database, there was training on how to use the database. When you buy a new database, you must provide training and tell people how to use it. This is the same for RDMS. Training should be diversified, such as data literacy, data retrieval, and data analysis. (Interview 35)

Additionally, to ensure researchers can easily access "useful information to develop their own RDM plan" (Interview 7), the library should at least provide "easily understandable data management guidelines" in the form of manuals, brochures, and webpages, from which "researchers can quickly capture information needed" (Interview 38). In fact, it was perceived during the data analysis that establishing RDM training courses and providing explicit guidelines should not be difficult and challenging. UYL could learn from many other university libraries

around the world that have been developed and offer RDM planning services. Some of the RDM guidelines are freely available on the Internet.

Research data collection

Collecting research data is important for any type of research investigation. The interview data revealed that the library can help participants access and gather a variety of research data types. The data analysis pointed to three RDMS requirements:

- Access to open data resources
- Guidance on research data collection
- Institutional research data repository.

Several participants claimed that they are "happy and satisfied" with the current e-services, particularly the full-text access to academic databases, published papers (Interview 40) and electronic books (Interview 38). Some stated that they want "an easy method of obtaining open research data from the library website, just like how we access e-books from the library website" (Interview 2).

Moreover, UYL is expected to collect and curate "open data resources available from around the world" (Interview 2) and develop "a full catalogue of different types of research data" (Interview 37). Library staff should provide convenient assistance – as one interviewee stated, "telling me how to use the data available in the library" (Interview 4).

> I would like to have a comprehensive collection of research data available at the library. Additionally, I desire immediate accessibility to the research data based on [a] researcher's individual needs. Also, I am interested in exploring various methods of gathering open-access data. (Interview 2)

Data collection can be challenging. Some interview participants stated that UYL needs to be "more active in helping and guiding researchers to learn and gain more experience on how to collect data and store data" (Interview 38). This requirement was repeatedly raised by "novice researchers" and students, who often struggle at this research stage. UYL staff can step in when they need help and serve as their "second professor" (Interview 38), but the UYL services are "disappointing" (Interview 4).

Development of an institutional data repository was requested by several participants. "The data repository [development] should be included as a part of [the] library's RDMS strategy" (Interview 38). The ideal data repository should allow individual researchers to "establish their own private storage" (Interview 40), "curate and manage research data", as well as "sharing data among colleagues and members within a research team" (Interview 39). It should be "easy to operate all the time, especially when in the field collecting data" (Interview 40).

> I'd want the library [to] set up a system in which I can upload and save my data and I can check again and again what I've read. Regarding data storage, the library will be able to build the system where the researchers can store and share their research data. (Interview 39)

Research data processing and analysis

Collected research data need to be processed and analysed through a series of procedures requiring librarian support. Three requirements emerged from the analysis:

- Training on research data processing
- Access to data analysis applications
- Hands-on assistance with research data processing and analysis.

Several participants expressed their need for librarians' assistance regarding "how to conduct the data process systematically" (Interview 38) and "how to store and analyse data systematically" (Interview 39). Participants expect a reliable, secure, and easily accessible platform for data storage, as well as "systematic training" (Interview 4).

> I would like to receive training on how to collect, store, and prepare data in a systematic way. I'm looking for training that may be organised systematically for data activities related to research projects, such as data preservation, analysis, and storage. (Interview 39)

A very challenging requirement is to offer a variety of "software and applications that are easy to use for entering, transcribing, storing and analysing data" (Interview 8) because "some applications are quite expensive" (Interview 7) when considering the limited annual budget of UYL. In this case, it is "almost impossible for researchers to access and utilise [these applications]" (Interview 4).

Participants also considered that the library should "help with the data process and analysis procedures" (Interview 9) and "offer on-site guidance on how to use those data process applications".

Support from the university leadership was perceived as a critical factor and determinant for the success of RDMS.

> Because I do physics and material research, I need to use 3D software. I use graphic software for reproduction. In most cases, I learned and used the software by myself. I can't use the software as well as a real expert. It would be great if a librarian could teach me how to use the software properly. (Interview 6)

Research data preservation and sharing

The library plays a crucial role in facilitating and advancing the preservation of research data to ensure the accessibility and re-use of data. To enable the preservation and sharing of research data, the participants mentioned three RDM requirements:

- Long-term preservation of research data
- Promoting awareness about the importance of data preservation and sharing
- Policies and regulations to encourage research data sharing.

Although no data preservation services are provided at present, most of the researchers were very interested in receiving support from UYL in this regard. Particularly, some interviewees expressed expectations for RDMS that ensures "long-lasting data preservation" (Interview 36), enabling them to access research data for "long-term use" (Interview 17). RDMS should not only include "storing data" services (Interview 31) but also "services that make sure data remains accessible and usable over extended periods" (Interview 12, 31, 38). Even if it is "not possible to have a complete set of services in the short run – including building [a] university storage platform, toolset, and everything – at least the library could consider offering [a] guidance service" (Interview 6).

> There is no specific arrangement in the library. It would be better to have a good service for long-term preservation. As a library, there is no service related to data storage and preservation. I cannot provide [self] funding for storage every month. (Interview 6)

Additionally, the interview participants revealed that "most researchers keep their research data in their USB drive" and thus "it is a significant waste of useful data" (Interview 11, 18). Therefore, UYL should become active in "enhancing the robustness and reliability of data preservation through storage and maintenance methods" (Interview 17), and in "promoting awareness and educating researchers regarding research data preservation" (Interview 3).

> In terms of long-term preservation, the data is saved on USB. CDs are also utilised. I also utilise email. The software I use also saves data. I would like the library to provide a storage drive or an online space that can be used to backup, store, and preserve research data permanently. I hope the library develops a storage system that allows users to create their personal storage. (Interview 40)

Furthermore, participants noted that it is "difficult enough to share research data between the members of the research team" (Interview 19) because there are absolutely "no rules, policies, or guidelines for sharing research data" (Interview 25, 39). It would be unthinkably challenging to make cross-team, cross-school and even cross-university sharing. The interview participants believe that UYL should work with the university leadership to establish policies and regulations aiming to encourage the effective sharing of research data.

Discussion

The analysis of interview data revealed 12 RDMS requirements, which respond to the first research question (RQ1). Also, these requirements indicate directions for the remainder stages of research, as well as the development of RDMS at UYL. In response to the second research question (RQ2), the data analysis offered three implications:

Firstly, it was noticed during the analysis that significant barriers were hindering the development of RDMS as part of the university library services. These barriers include a lack of leadership support, insufficient financial support, and an absence of technological infrastructure. Barriers to the development of RDMS are reported in several studies (e.g., Birkbeck et al., 2022; Mushi, 2021). For instance, Birkbeck et al. (2022) point out that the lack of a capable library service team and the lack of necessary education are barriers hindering RDMS development in libraries. Also, Sinha et al. (2023) claim that limited technical competency and the lack of effective technological infrastructure severely reduced the performance and quality of RDMS services. Therefore, development needs to start from a basic level, possibly requiring the destruction of existing organisational

structure and workflows and the development of new ones. Thus, as suggested by Ismail et al. (2022) and Palsdottir (2021), it is important to study, understand and possibly mitigate barriers to RDMS development during the remainder research stages.

Secondly, RDMS should be offered along the lifecycle of a research project (Tunmibi and Ajokotola, 2024). According to the analysis, RDMS can be provided in four main stages of a project lifecycle: RDM planning, research data collection, research data processing and analysis, and research data presentation and sharing. This finding is consistent with other research studies, although some report different lifecycle stages (e.g., Amanullah and Abrizah, 2023; Tenopir et al., 2012). The four lifecycle stages identified in this study are more likely to be useful and appropriate for RDMS development in UYL.

Thirdly, the user requirements point to three roles for library involvement in RDMS. In the *participative role*, the RDMS librarian becomes a member of a research team and provides related services by working side-by-side with other researchers in the team (Federer, 2016; Gowen and Meier, 2020). The *educational and consultative role* requires RDMS librarians to provide several types of education programs, as well as timely consultations and advice at the request of researchers (Joo and Schmidt, 2021; Tenopir et al., 2019;). In the *facilitative role*, UYL develops technological infrastructure that enables access to a comprehensive set of data resources, online tools, and applications (Masinda et al., 2021). UYL should also participate in the development of university policies and work with external organisations, e.g., publishers, public and private data repositories (Kim, 2022). The identification of these different roles could provide a useful basis for designing RDMS in practice.

Moreover, to respond to the third research question (RQ3), a tentative action plan can be developed and includes three stages. The first should focus on ensuring external conditions are met for RDMS development, which usually includes soliciting support, establishing policies and guidelines, and building technological infrastructure. This stage requires the library to work closely with the university leadership and win over their support. Based on the first stage, the second stage is more concerned with the development and formation of RDMS, typically involving building a holistic service architecture, restructuring the existing flow of library services, creating new service roles and responsibilities, and removing barriers that may hinder implementation. Stage three facilitates the interaction and negotiation among UYL, university leadership, and the users of RDMS, aiming to continuously improve the proficiency and quality of RDMS provision.

Conclusions

This report introduced a research study aiming to develop RDMS in university libraries in Myanmar. Specifically, the paper focused on the identification of RDMS user requirements and pointed to a total of 12 requirements. Moreover, through the data analysis, the requirements identified were categorised into four themes, which can be conceptualised as four RDM stages along a research project life cycle. These themes are: RDM planning, research data collection, research data processing and analysis, and research data preservation and sharing.

These requirements have theoretical and applied value. From a theoretical perspective, first, the identification of user requirements adds to the understanding and literature of RDMS development, and its interaction with users. Second, it is evident that the research project lifecycle could serve as a conceptual baseline for the investigation, design and development of RDMS. Third, university libraries can assume three roles in RDMS: a participative role, an educational and consultative role, and a facilitative role. From a practical perspective, the requirements identified could be used as an evidence base for service formulation. RDMS development at UYL could follow three successive stages: developing supportive internal and external conditions, building RDMS and service architecture, and revising and improving RDMS.

The research findings offer important indications for the RDMS development at UYL. As shown, there exist barriers and constraints, which hinder and prevent UYL's RDMS development. This finding was not anticipated in the original research design. It is reasonable to speculate that overlooking these barriers can cause the total failure of the RDMS development project. Therefore, the next research stage will concentrate on identifying and comprehending the barriers and devising effective mitigating strategies.

This study carries important values that are pertinent not only to university libraries in Myanmar, but also to those in the developing world. Moreover, the findings are useful for university libraries across the globe in the development of similar kinds of research support services.

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